Problem Sheets *Six & Seven*

A complete academic reference covering probability mass functions, cumulative distribution functions, probability density functions, expected value, and variance — with fully worked solutions, theorem proofs, examples, and examination guidance.

TOPICS	QUESTIONS	PREPARED
PMF · PDF · CDF · E[X] · Var(X)	9 fully worked solutions	January 2025

READING GUIDE

How to Use This Document

Every annotation block is colour-coded by function. Learn the system once; it applies throughout.

- Definition / Concept
- Theorem & Proof
- Formula / Key Result
- Worked Example
- Exam Warning
- Tip / Strategy

CHAPTER I

Foundational Concepts

The building blocks every subsequent theorem and solution depends on.

Random Variables

DEFINITION — RANDOM VARIABLE

A **random variable** X is a function $X : \Omega \to \mathbb{R}$ that assigns a real number to each outcome ω in a sample space Ω. The randomness lives in the experiment; the variable just translates outcomes into numbers we can do arithmetic with.

Discrete if it takes countably many values (e.g. integers); *continuous* if it takes values in an interval of $\mathbb{R}$.

EXAMPLE — WHAT "RANDOM VARIABLE" ACTUALLY MEANS

Roll a die. The sample space is $\Omega = \{1, 2, 3, 4, 5, 6\}$. Define $X(\omega) = \omega$. This maps each face to its numeric value — so X is a discrete RV. Now define $Y(\omega) = \mathbf{1}[\omega \text{ is even}]$, which maps even rolls to 1 and odd rolls to 0. That is also a valid discrete RV. The same experiment supports many different random variables.

The Probability Mass Function (Discrete)

DEFINITION — PROBABILITY MASS FUNCTION (PMF)

For a discrete RV X, the **probability mass function** is the function $p : \mathbb{R} \to [0, 1]$ defined by:

$$p(x) = P(X = x)$$

It assigns to each possible value x the probability that X equals exactly that value. The set of values where $p(x) > 0$ is called the *support* of X.

Validity conditions — both must hold simultaneously:

- **Non-negativity:** $p(x) \geq 0$ for every $x \in \mathbb{R}$
- **Normalisation:** $\sum_{\text{all } x} p(x) = 1$

COMMON ERROR

Normalisation does not guarantee non-negativity, and non-negativity does not guarantee normalisation. You must verify *both* conditions independently. In PS6 Q1b, the sum of $\frac{1}{5}(2x - 3)$

over $x = 0 \dots 5$ actually does equal 1 — yet the PMF is still invalid because $p(0) = -0.6 < 0$. Checking only the sum would give a false pass.

The Probability Density Function (Continuous)

DEFINITION — PROBABILITY DENSITY FUNCTION (PDF)

For a continuous RV X, the **probability density function** $f : \mathbb{R} \to [0, \infty)$ is the function such that probabilities of intervals are computed by integration:

$$P(a \leq X \leq b) = \int_a^b f(x)\, dx$$

Validity conditions:

- $f(x) \geq 0$ for all x
- $\displaystyle\int_{-\infty}^{\infty} f(x)\, dx = 1$

Critical: $f(x)$ is *not* a probability itself — it is a *density*. It can exceed 1. The probability of any single point is always zero: $P(X = x) = 0$.

EXAMPLE — WHY F(X) CAN EXCEED 1

Consider $f(x) = 4x$ on $[0, 0.5]$. At $x = 0.4$, we have $f(0.4) = 1.6 > 1$. This is perfectly valid: the area under the curve from 0 to 0.5 equals $\int_0^{0.5} 4x\, dx = [2x^2]_0^{0.5} = 2(0.25) = 0.5$. But wait — that only equals 0.5, not 1. So this specific function is actually not a valid PDF. To fix it, use $f(x) = 4x$ on $[0, 1/\sqrt{2}]$ or normalise it. The point stands: density values greater than 1 are fine as long as the integral equals 1.

The Cumulative Distribution Function

DEFINITION — CDF

The **cumulative distribution function** of any RV X (discrete or continuous) is:

$$F(x) = P(X \leq x)$$

For discrete X: $F(x) = \sum_{t \leq x} p(t)$ — a right-continuous step function.

For continuous X: $F(x) = \int_{-\infty}^{x} f(t)\,dt$ — a smooth, non-decreasing function.

Universal properties of every CDF:

- $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to+\infty} F(x) = 1$
- F is non-decreasing: $a \le b \Rightarrow F(a) \le F(b)$
- For the discrete case, F is a staircase: it is constant between support points and jumps at each support point.

CHAPTER II

Theorems, Explained & Applied

Each theorem is stated, proved intuitively, and illustrated with a concrete example showing exactly when and why you use it.

Theorem 1 — The Geometric Series

THEOREM

Geometric Series Sum

For any real number r with $|r| < 1$:

$$\sum_{i=0}^{\infty} r^i = \frac{1}{1-r}$$

More generally, with leading coefficient a:

$$\sum_{i=0}^{\infty} ar^i = \frac{a}{1-r}$$

Why does this work? The partial sum $S_n = 1 + r + r^2 + \cdots + r^n = \frac{1-r^{n+1}}{1-r}$. As $n \to \infty$, if $|r| < 1$, then $r^{n+1} \to 0$, leaving $\frac{1}{1-r}$. The condition $|r| < 1$ is essential — the series diverges if $|r| \geq 1$.

USE CASE — VERIFYING THE GEOMETRIC PMF (PS6 Q1A)

Given $p(x) = 0.25 \cdot (0.75)^x$ for $x = 0, 1, 2, \ldots$

We need to check whether $\sum_{x=0}^{\infty} p(x) = 1$. Here $a = 0.25$ and $r = 0.75$. Since $0.75 < 1$, the theorem applies:

$$\sum_{x=0}^{\infty} 0.25 \cdot (0.75)^x = \frac{0.25}{1 - 0.75} = \frac{0.25}{0.25} = 1 \checkmark$$

Without this theorem, you would need to sum infinitely many terms — impossible by hand. The theorem reduces it to a single division.

Pattern: Any PMF of the form $p(x) = (1 - r)\, r^x$ for $x = 0, 1, 2, \ldots$ with $0 < r < 1$ is automatically normalised by this theorem. It defines the Geometric distribution.

WHEN TO APPLY THIS THEOREM

Whenever a PMF has an infinite support $\{0, 1, 2, \ldots\}$ and the formula involves a constant raised to the power x. Identify a (the coefficient) and r (the base), verify $|r| < 1$, and apply the formula directly. Never attempt to sum these terms numerically.

Theorem 2 — Recovering the PMF from the CDF

THEOREM

Jump-Size Extraction

For a discrete RV with CDF F, the PMF value at any support point x_0 equals the magnitude of the jump in F at that point:

$$p(x_0) = F(x_0) - \lim_{x \nearrow x_0} F(x) = F(x_0) - F(x_0^-)$$

where $F(x_0^-)$ denotes the left-hand limit — the value of F approached from just below x_0.

The intuition is simple: the CDF accumulates all probability mass up to and including x. When it jumps at a point, that jump is precisely the probability that X equals that point. Between support points, the CDF is flat — no new probability is accumulating.

USE CASE — PS6 Q4 IN FULL DETAIL

Given the CDF with breakpoints at $x = 0, 1, 2, 3, 4$. Think of the CDF as a staircase — each step-up is a point mass. Here is the reading method:

LOCATION X_0	$F(X_0^-)$ — VALUE JUST BEFORE	$F(X_0)$ — VALUE AT	$P(X_0)$ = JUMP SIZE
0	0 (flat at 0 for $x < 0$)	$1/2$	$1/2 - 0 = 1/2$
1	$1/2$ (flat on $[0, 1)$)	$3/5$	$3/5 - 1/2 = 1/10$
2	$3/5$ (flat on $[1, 2)$)	$4/5$	$4/5 - 3/5 = 1/5$
3	$4/5$ (flat on $[2, 3)$)	$9/10$	$9/10 - 8/10 = 1/10$
4	$9/10$ (flat on $[3, 4)$)	1	$1 - 9/10 = 1/10$

Verification: $\frac{1}{2} + \frac{1}{10} + \frac{2}{10} + \frac{1}{10} + \frac{1}{10} = \frac{10}{10} = 1$ ✓. The staircase has five steps; we extract five PMF values, one per step.

EXAM TRAP — THE MOST COMMON CDF ERROR

Students write $p(x) = F(x)$ instead of $p(x) = F(x) - F(x^-)$. That would give $p(1) = 3/5$ instead of the correct $1/10$. The CDF is cumulative — it includes all mass up to and including that point. You must subtract what was already accumulated before arriving at x.

Theorem 3 — Expected Value

THEOREM

Expected Value (Mean)

The **expected value** $E[X]$ (also written μ) is the probability-weighted average of all possible values of X.

$$E[X] = \begin{cases} \displaystyle\sum_{\text{all } x} x \cdot p(x) & \text{(discrete)} \\ \displaystyle\int_{-\infty}^{\infty} x \cdot f(x)\, dx & \text{(continuous)} \end{cases}$$

For any function $g(X)$, the **Law of the Unconscious Statistician** gives:

$$E[g(X)] = \sum_x g(x)\, p(x) \quad \text{or} \quad \int_{-\infty}^{\infty} g(x)\, f(x)\, dx$$

This is used directly when computing $E[X^2]$ by setting $g(x) = x^2$.

Think of $E[X]$ as a physical centre of mass. If you placed weights $p(x)$ at positions x on a number line, the point at which the line balances is $E[X]$. High-probability values pull the centre toward themselves; low-probability values in the tail contribute little.

USE CASE — PS7 Q3: COMPUTING E[X] WHEN PROBABILITIES ARE INCOMPLETE

X = hits for a softball player. We have $P(X = 1) = 0.3$, $P(X = 2) = 0.2$, $P(X = 0) = 3P(X = 3)$. We cannot compute $E[X]$ until we know all four probabilities. Let $P(X = 3) = p$. Then $P(X = 0) = 3p$. Normalisation gives $3p + 0.3 + 0.2 + p = 1 \Rightarrow p = 0.125$. Now:

X	$P(X)$	$X \cdot P(X)$	$X^2 \cdot P(X)$
0	0.375	0	0
1	0.300	0.300	0.300
2	0.200	0.400	0.800
3	0.125	0.375	1.125
Sum	1.000	**1.075**	**2.225**

$E[X] = 1.075$ hits. On a long run of games, the player averages about 1.075 hits per 3 bats.

Theorem 4 — Variance and the Shortcut Formula

THEOREM

Variance & the Computational Shortcut

The **variance** $\text{Var}(X) = \sigma^2$ measures the average squared deviation from the mean:

$$\text{Var}(X) = E\left[(X - \mu)^2\right]$$

Expanding the square and applying linearity, this simplifies to the **shortcut formula**:

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

Proof of the shortcut:

$$\begin{aligned} E[(X-\mu)^2] &= E[X^2 - 2\mu X + \mu^2] \\ &= E[X^2] - 2\mu E[X] + \mu^2 \\ &= E[X^2] - 2\mu^2 + \mu^2 \\ &= E[X^2] - \mu^2 \end{aligned}$$

Variance answers the question: how spread out is the distribution? A variance of 0 means X is a constant. Large variance means values are scattered far from the mean. The standard deviation $\sigma = \sqrt{\text{Var}(X)}$ *restores the original units.*

USE CASE — WHY THE SHORTCUT IS ALWAYS PREFERRED

For PS7 Q3, computing $\textbf{Var}(X)$ from the definition would require: $(0-1.075)^2(0.375) + (1-1.075)^2(0.3) + (2-1.075)^2(0.2) + (3-1.075)^2(0.125)$. Each term involves squaring a decimal, then multiplying — messy and error-prone. The shortcut gives: $E[X^2] - (E[X])^2 = 2.225 - (1.075)^2 = 2.225 - 1.155625 = 1.069375$. The table-based approach builds $E[X^2]$ at the same time as $E[X]$ — no extra work.

EXAM TRAP — SQUARING THE WRONG THING

The formula is $E[X^2] - \left(E[X]\right)^2$. Students sometimes compute $E[X^2 - E[X]^2]$ instead, or they forget to square $E[X]$ and write $E[X^2] - E[X]$. These produce wrong numbers. Always: compute $\mu = E[X]$

as a decimal, square it, then subtract from $E[X^2]$.

Theorem 5 — Linearity of Expectation

THEOREM

Linearity of Expectation & Variance Scaling

For constants $a, b \in \mathbb{R}$ and a random variable X:

$$E[aX + b] = a\,E[X] + b$$

$$\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$$

For **independent** random variables X and Y:

$$E[X \pm Y] = E[X] \pm E[Y]$$

$$\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$$

Note the critical asymmetry: the sign of $\pm$ does not affect variance — whether you add or subtract independent variables, the variances always *add*.

Linearity of expectation is perhaps the most powerful tool in all of probability theory. It holds regardless of dependence for expectations, and requires independence only for the variance result. Adding a constant b shifts every outcome equally — it shifts the mean but cannot spread the distribution. Multiplying by a stretches all deviations from the mean by |a|, so variance scales by a^2.

USE CASE — PS7 Q4: THE POOL COMPANY (COMPOUND RANDOM VARIABLE)

Here $X = 2 + J - C$ where J = new contracts received and C = contracts completed, with $J \perp C$ (given independent).

Expectation: $E[X] = E[2 + J - C] = 2 + E[J] - E[C]$. We compute $E[J] = 0.1(1) + 0.2(2) + 0.5(3) + 0.2(4) = 2.8$ and $E[C] = 0.5(2) + 0.5(3) = 2.5$. Therefore $E[X] = 2 + 2.8 - 2.5 = 2.3$.

Variance: The constant 2 contributes zero variance. Because $J \perp C$: $\text{Var}(X) = \text{Var}(J) + \text{Var}(C)$. We find $\text{Var}(J) = E[J^2] - (E[J])^2 = 8.6 - 7.84 = 0.76$ and $\text{Var}(C) = 6.5 - 6.25 = 0.25$. So $\text{Var}(X) = 0.76 + 0.25 = 1.01$.

The key insight: we never needed to enumerate every possible value of X explicitly. Linearity let us decompose a complex RV into simpler independent parts and handle each separately.

DEEPER EXAMPLE — WHY VAR(X - Y) = VAR(X) + VAR(Y) FOR INDEPENDENT X, Y

This surprises many students. Intuitively: subtracting Y from X does not reduce uncertainty — if anything, the result is harder to predict than either alone. Formally: $\text{Var}(X - Y) = \text{Var}(X) + (-1)^2\text{Var}(Y) = \text{Var}(X) + \text{Var}(Y)$. The $(-1)^2 = 1$ is why the sign disappears. Independence is required; for dependent variables, a covariance term appears: $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2\,\text{Cov}(X, Y)$.

Theorem 6 — Computing Probabilities via Integration

THEOREM

Probabilities as Areas Under the Density Curve

For a continuous RV X with PDF f and CDF F:

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx = F(b) - F(a)$$

The CDF and PDF are related by the Fundamental Theorem of Calculus:

$$F(x) = \int_{-\infty}^{x} f(t)\,dt \qquad \Longleftrightarrow \qquad f(x) = F'(x)$$

Because $P(X = c) = 0$ for any constant c, all four inequalities are interchangeable: $P(X < b) = P(X \leq b) = F(b)$. This is a property of continuous distributions only.

The PDF does not give probabilities directly — it gives probability per unit of x. A tall spike of f means X is very likely to be near that region. The total area is always 1, representing certainty that X takes some value.

USE CASE — PS7 Q1: FINDING A THRESHOLD A

Given $f(x) = 2x$ on $[0, 1]$, we want α such that $P(X < \alpha) = 3 \cdot P(X \geq \alpha)$. The two events partition the real line, so they sum to 1. Let $p = P(X < \alpha)$; then $P(X \geq \alpha) = 1 - p$. The condition says $p = 3(1 - p)$, giving $p = 3/4$. Now solve $\int_0^\alpha 2x\,dx = 3/4$, i.e., $\alpha^2 = 3/4$, giving $\alpha = \frac{\sqrt{3}}{2}$. Integration is the mechanism; the algebra comes from setting up the condition correctly.

USE CASE — PS7 Q2: FINDING THE MEDIAN OF AN EXPONENTIAL DISTRIBUTION

With $f(x) = e^{-x}$, the CDF is $F(t) = 1 - e^{-t}$. Setting $F(t) = 0.5$ means $1 - e^{-t} = 0.5$, so $e^{-t} = 0.5$, hence $t = \ln 2 \approx 0.693$. The median is not the mean here: $E[X] = 1$ (for Exp(1)), while the median is $\ln 2 \approx 0.693$. The distribution is right-skewed — the mean is pulled rightward by the long tail, while the median is the equal-split point.

Theorem 7 — Determining Unknown Constants by Normalisation

THEOREM

The Normalisation Principle

If a PMF or PDF contains an unknown constant k, it is uniquely determined by the requirement that probabilities sum (or integrate) to 1:

$$\text{Discrete: } \sum_x \frac{k}{\phi(x)} = 1 \;\Rightarrow\; k = \frac{1}{\sum_x \frac{1}{\phi(x)}}$$

$$\text{Continuous: } \int_{\text{support}} k\,g(x)\,dx = 1 \;\Rightarrow\; k = \frac{1}{\int_{\text{support}} g(x)\,dx}$$

This works because exactly one value of k satisfies both (i) the given functional form and (ii) the total-probability axiom.

Think of k as a scaling factor that "pushes" an unnormalised shape into a valid probability distribution. The shape of the distribution (which outcomes are relatively more or less likely) is encoded in the function body; k merely rescales it so the total is 1.

USE CASE — PS6 Q2: FACTORIAL PMF

Given $p(x) = k/x!$ for $x = 0, 1, 2, 3, 4$. The sum is $k\sum_{x=0}^{4} \frac{1}{x!} = k(1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24}) = k \cdot \frac{65}{24}$. Setting this equal to 1 gives $k = \frac{24}{65}$. Notice that if the sum extended to infinity, we would get $k \cdot e = 1$, so $k = e^{-1}$ — the Poisson(1) normalising constant. Truncating at $x = 4$ changes the value of k slightly.

USE CASE — PS7 Q5: POLYNOMIAL PDF

Given $f(x) = \frac{3}{4}(kx - x^2)$ on $[0, 2]$. Integrate: $\frac{3}{4}\int_0^2 (kx - x^2)\,dx = \frac{3}{4}[k \cdot \frac{x^2}{2} - \frac{x^3}{3}]_0^2 = \frac{3}{4}(2k - \frac{8}{3})$. Set equal to 1: $2k - \frac{8}{3} = \frac{4}{3} \Rightarrow 2k = 4 \Rightarrow k = 2$. The constant $\frac{3}{4}$ was pre-included in the design of the function; $k = 2$ completes it. With $k = 2$, $f(x) = \frac{3}{4}x(2 - x)$, which is a smooth bell shape on $[0, 2]$.

Theorem 8 — Combinatorial Identities in PMFs

THEOREM

Binomial Coefficient Convention

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} \quad \text{for } 0 \leq k \leq n$$

By convention (and by the vanishing of the factorial formula): $\binom{n}{k} = 0$ whenever $k < 0$ or $k > n$. This is not just a notational trick — it reflects the fact that there are zero ways to choose a negative number of items from a set, or more items than exist.

The **Vandermonde identity** states $\sum_{k=0}^{r} \binom{m}{k}\binom{n}{r-k} = \binom{m+n}{r}$, which is used to verify PMFs involving products of binomial coefficients.

USE CASE — PS6 Q1C: PRODUCT OF BINOMIALS

The function $p(x) = \frac{1}{36}\binom{5}{x}\binom{4}{x-2}$ has support $x \in \{0, 1, 2\}$. At $x = 0$: $\binom{4}{-2} = 0$. At $x = 1$: $\binom{4}{-1} = 0$. Only at $x = 2$ do both coefficients exist: $\binom{5}{2}\binom{4}{0} = 10 \cdot 1 = 10$, giving $p(2) = 10/36$. Since $10/36 \neq 1$, this is not a valid PMF. The "natural" support of this hypergeometric-type expression extends to $x \in \{2, 3, 4, 5, 6\}$, but the given support cuts it off, leaving most mass undefined.

CHAPTER III

Examination Traps & Errors

These are the precise points where marks are lost. Each is documented with the incorrect reasoning and the correct alternative.

TRAP 1 — CHECKING ONLY NORMALISATION, NOT NON-NEGATIVITY

Incorrect reasoning: "The sum equals 1, so this must be a valid PMF."
Why it fails: A function can sum to 1 while taking negative values. Negative probabilities are logically impossible — they cannot represent frequencies of events. Always check non-negativity first; if it fails, stop immediately.

Affected question: PS6 Q1b, where $p(0) = -3/5$ invalidates the function despite the sum being correct.

TRAP 2 — TREATING $\binom{N}{K}$ AS NON-ZERO FOR NEGATIVE K

Incorrect reasoning: Computing $\binom{4}{-2}$ as some positive fraction.
Why it fails: The binomial coefficient $\binom{n}{k}$ for integer $k < 0$ is identically zero by the combinatorial definition. There is no "formula" to apply — the result is simply 0. This directly affects which values of x contribute to the PMF sum.

Affected question: PS6 Q1c at $x = 0$ and $x = 1$.

TRAP 3 — P(X < N) VERSUS P(X ≤ N) FOR DISCRETE VARIABLES

Incorrect reasoning: Treating these as equivalent.
Why it fails: For discrete X, these differ by exactly $p(n)$:

$$P(X < n) = P(X \leq n-1) = \sum_{x \leq n-1} p(x)$$

The value $x = n$ is included in $P(X \leq n)$ but excluded from $P(X < n)$.
For continuous X only: $P(X < n) = P(X \leq n)$ because $P(X = n) = 0$.
Affected question: PS6 Q2b: $P(X < 2) = p(0) + p(1)$, not $p(0) + p(1) + p(2)$.

TRAP 4 — COMPUTING E[X] OR VAR(X) BEFORE FINDING K

Incorrect reasoning: Plugging into the expectation formula while k is still unknown.
Why it fails: Every probability value $p(x)$ or $f(x)$ depends on k. Computing sums or integrals without it produces answers that are off by some unknown scaling factor.
Rule: In any problem containing an unspecified constant, step one is always to find that constant. No other computation is valid until this is done.

TRAP 5 — THE VARIANCE SQUARING ERROR

Incorrect reasoning: Writing $\mathrm{Var}(X) = E[X^2] - E[X]^2$ but then computing $E[X^2 - E[X]^2]$ by folding the subtraction inside the expectation operator.
Why it fails: $E[X^2] - (E[X])^2 \neq E[X^2 - E[X]^2]$ unless $E[X] = 0$. The term $(E[X])^2$ is a constant — it must be taken outside the expectation and subtracted numerically. Compute $\mu = E[X]$ as a number, then compute μ^2 as a number, then subtract from $E[X^2]$.

TRAP 6 — USING THE WRONG INTEGRATION LIMITS

Incorrect reasoning: Integrating from $-\infty$ to ∞ when the PDF has a finite support.
Why it matters: This is not technically wrong (since $f(x) = 0$ outside the support), but in practice it leads to errors in the antiderivative evaluation: forgetting to substitute the lower bound, or evaluating at 0 when the support starts at 2, for instance.
Rule: Always write the integral with the explicit support bounds as limits, e.g. $\int_0^2$ for PS7 Q5, not $\int_{-\infty}^{\infty}$
.

TRAP 7 — FORGETTING A MISSING PROBABILITY (PS7 Q3)

Incorrect reasoning: Computing $E[X] = 0 \cdot P(X = 0) + 1(0.3) + 2(0.2) + 3(0.125)$ without first establishing $P(X = 0)$.
Why it fails: $P(X = 0)$ is not given — it is implicitly defined via $P(X = 0) = 3P(X = 3)$. If you insert a wrong value (e.g. treating $P(X = 0)$ as some default), the sum will not equal 1 and $E[X]$ will be wrong. The missing probability must be recovered algebraically before any computation.

CHAPTER IV

Problem Sheet 6 — Worked Solutions

PMF validity, unknown constants, bar charts, CDF extraction. Every step shown in full.

QUESTION 1(A) *Is the following a valid probability mass function?*

$p(x) = 0.25\,(0.75)^x$ for $x = 0, 1, 2, \ldots$ and $p(x) = 0$ otherwise.

STRATEGY

Recognise the geometric PMF pattern. Apply the geometric series theorem directly — do not attempt partial sums. Two conditions must both hold.

1. **Non-negativity.** We have $0.25 > 0$ and $(0.75)^x > 0$ for every integer $x \geq 0$, since any positive base raised to any real power remains positive. Therefore $p(x) > 0$ for all x in the support. ✓

2. **Normalisation.** We must show $\sum_{x=0}^{\infty} 0.25 \cdot (0.75)^x = 1$. This is a geometric series with $a = 0.25$ and ratio $r = 0.75$. Since $|r| = 0.75 < 1$, the theorem applies:

$$\sum_{x=0}^{\infty} 0.25 \cdot (0.75)^x = \frac{0.25}{1 - 0.75} = \frac{0.25}{0.25} = 1 \checkmark$$

CONCLUSION

Valid PMF. Both conditions are satisfied. This is the Geometric(0.25) distribution — the probability of achieving the first success on the $(x+1)$th Bernoulli trial with success probability 0.25.

QUESTION 1(B) *Is the following a valid probability mass function?*

$p(x) = \frac{1}{5}(2x - 3)$ for $x = 0, 1, 2, 3, 4, 5$ and $p(x) = 0$ otherwise.

1. **Check non-negativity at each support point.** Compute $p(x)$ explicitly:

X	$2X - 3$	$P(X) = \frac{1}{5}(2X - 3)$	SIGN
0	-3	$-3/5 = -0.6$	**Negative** ✗
1	-1	$-1/5 = -0.2$	**Negative** ✗
2	1	$1/5 = 0.2$	**Positive** ✓
3	3	$3/5 = 0.6$	**Positive** ✓
4	5	$5/5 = 1.0$	**Positive** ✓
5	7	$7/5 = 1.4$	**Exceeds 1 (still valid if sum = 1)**

At $x = 0$, $p(0) = -0.6 < 0$. Non-negativity fails. Stop here.

CONCLUSION

Not a valid PMF. $p(0) = -3/5 < 0$ and $p(1) = -1/5 < 0$. The non-negativity axiom is violated. No need to check the sum — once any value is negative, the function cannot represent a probability distribution.

QUESTION 1(C) *Is the following a valid probability mass function?*

$p(x) = \dfrac{\binom{5}{x}\binom{4}{x-2}}{36}$ for $x = 0, 1, 2$ and $p(x) = 0$ otherwise.

1 **Evaluate each term, applying the zero-convention for negative arguments.**

X	$\binom{5}{X}$	$\binom{4}{X-2}$ — ARGUMENT	VALUE	$P(X)$
0	1	$\binom{4}{-2}$ — negative	0	0
1	5	$\binom{4}{-1}$ — negative	0	0
2	$\binom{5}{2} = 10$	$\binom{4}{0} = 1$	10	$10/36$

2 **Non-negativity.** All values are ≥ 0. ✓

3 **Normalisation check:**

$$\sum_{x=0}^{2} p(x) = 0 + 0 + \frac{10}{36} = \frac{10}{36} \approx 0.278 \neq 1$$

The sum is less than 1 — normalisation fails.

CONCLUSION

Not a valid PMF. Although all values are non-negative, the sum over the given support equals only $10/36 \approx 0.278$, far below 1. The chosen support $\{0, 1, 2\}$ is too restrictive — the expression would yield non-zero values for $x \in \{2, 3, 4, 5, 6\}$, but those are excluded by the problem definition, leaving the distribution unnormalised.

QUESTION 2 *Find k such that the function is a PMF, then compute $P(X < 2)$.*

$p(x) = k/x!$ for $x = 0, 1, 2, 3, 4$ and $p(x) = 0$ otherwise.

1. **Compute the factorial values.** The five terms in the sum:

X	$X!$	$1/X!$	AS FRACTION OVER 24
0	1	1	24/24
1	1	1	24/24
2	2	1/2	12/24
3	6	1/6	4/24
4	24	1/24	1/24

2. **Apply the normalisation condition $\sum p(x) = 1$:**

$$k\left(\frac{24 + 24 + 12 + 4 + 1}{24}\right) = k \cdot \frac{65}{24} = 1 \implies k = \frac{24}{65}$$

3. **Compute $P(X < 2)$.** Since X is discrete, $P(X < 2) = P(X = 0) + P(X = 1)$. The value $x = 2$ is strictly excluded.

$$P(X < 2) = \frac{k}{0!} + \frac{k}{1!} = k + k = 2k = \frac{48}{65} \approx 0.738$$

4. **Verification.** The complete PMF:

$$p(0) = \frac{24}{65}, \quad p(1) = \frac{24}{65}, \quad p(2) = \frac{12}{65}, \quad p(3) = \frac{4}{65}, \quad p(4) = \frac{1}{65}$$

Sum: $(24 + 24 + 12 + 4 + 1)/65 = 65/65 = 1$ ✓

ANSWERS

(a) $k = \frac{24}{65} \approx 0.369$

(b) $P(X < 2) = \frac{48}{65} \approx 0.738$

CONNECTION TO THE POISSON DISTRIBUTION

If the support were all non-negative integers, the normalising constant would be $k = e^{-1}$, making this a Poisson(1) PMF: $p(x) = e^{-1}/x!$. Truncating at $x = 4$ changes k to $24/65$ — slightly larger than $e^{-1} \approx 0.368$, because the missing tail probability from $x \geq 5$ must be redistributed.

QUESTION 3 *Plot the bar graph for the given PMF.*

$p(x) = \frac{12}{25x}$ for $x = 1, 2, 3, 4$ and $p(x) = 0$ otherwise.

1. **Compute all values.**

X	COMPUTATION	$P(X)$	DECIMAL
1	$12/(25 \times 1)$	$12/25$	0.480

X	COMPUTATION	$P(X)$	DECIMAL
2	$12/(25 \times 2)$	$6/25$	0.240
3	$12/(25 \times 3)$	$4/25$	0.160
4	$12/(25 \times 4)$	$3/25$	0.120

2 **Verify normalisation:**

$$\frac{12+6+4+3}{25} = \frac{25}{25} = 1 \checkmark$$

This confirms it is a valid PMF. The constant 12 was chosen precisely to make this work.

3 **Bar chart description.** Four vertical bars at $x \in \{1, 2, 3, 4\}$. Heights: 0.48, 0.24, 0.16, 0.12 respectively. The distribution is strictly decreasing — each bar is roughly half the previous (the harmonic $1/x$ decay). The y-axis runs from 0 to 0.5. Bars do not touch, emphasising the discrete nature of the variable. The distribution is right-skewed with most mass at $x = 1$.

BAR GRAPH DATA

$p(1) = 0.480$, $p(2) = 0.240$, $p(3) = 0.160$, $p(4) = 0.120$. All positive, sum = 1.000. The PMF is harmonic-decay shaped.

QUESTION 4 *Derive the PMF from the given CDF.*

$F(x) = 0$ for $x < 0$; $\frac{1}{2}$ for $0 \le x < 1$; $\frac{3}{5}$ for $1 \le x < 2$; $\frac{4}{5}$ for $2 \le x < 3$; $\frac{9}{10}$ for $3 \le x < 4$; 1 for $x \ge 4$.

METHOD

Identify the jump points — the values of x at which F changes from one constant to the next. At each jump point x_0, the PMF value is the size of the jump: $p(x_0) = F(x_0) - F(x_0^-)$.

1 $x = 0$: F transitions from 0 to 1/2.

$$p(0) = \frac{1}{2} - 0 = \frac{1}{2}$$

2. $x = 1$: F transitions from 1/2 to 3/5.

$$p(1) = \frac{3}{5} - \frac{1}{2} = \frac{6}{10} - \frac{5}{10} = \frac{1}{10}$$

3. $x = 2$: F transitions from 3/5 to 4/5.

$$p(2) = \frac{4}{5} - \frac{3}{5} = \frac{1}{5}$$

4. $x = 3$: F transitions from 4/5 to 9/10.

$$p(3) = \frac{9}{10} - \frac{8}{10} = \frac{1}{10}$$

5. $x = 4$: F transitions from 9/10 to 1.

$$p(4) = 1 - \frac{9}{10} = \frac{1}{10}$$

6. **Verify:**

$$\frac{1}{2} + \frac{1}{10} + \frac{2}{10} + \frac{1}{10} + \frac{1}{10} = \frac{5+1+2+1+1}{10} = \frac{10}{10} = 1 \checkmark$$

RECOVERED PMF

X	0	1	2	3	4
$p(x)$	$1/2$	$1/10$	$1/5$	$1/10$	$1/10$

Problem Sheet 7 — Worked Solutions

Continuous distributions, thresholds, expectation and variance. Every integral evaluated in detail.

QUESTION 1 *Find α such that $P(X < \alpha) = 3\,P(X \geq \alpha)$.*

$f(x) = 2x$ for $0 \leq x \leq 1$ and $f(x) = 0$ otherwise.

1. **Verify validity of the PDF.** $f(x) = 2x \geq 0$ on $[0, 1]$. $\int_0^1 2x\,dx = [x^2]_0^1 = 1$. ✓

2. **Recognise the partition.** The events $\{X < \alpha\}$ and $\{X \geq \alpha\}$ are mutually exclusive and exhaustive, so their probabilities sum to 1. Let $p = P(X < \alpha)$. Then $P(X \geq \alpha) = 1 - p$. The condition $p = 3(1-p)$ gives $4p = 3$, hence $p = 3/4$.

3. **Solve $P(X < \alpha) = 3/4$ via integration.**

$$\int_0^\alpha 2x\,dx = \left[x^2\right]_0^\alpha = \alpha^2 = \frac{3}{4}$$

$$\alpha = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2} \approx 0.866$$

4. **Reject the negative root.** $\alpha = -\sqrt{3}/2$ lies outside the support $[0, 1]$ and is discarded. The constraint $\alpha \in [0, 1]$ uniquely selects the positive root.

5. **Verification.** $P(X < 0.866) = (0.866)^2 = 0.75$. $P(X \geq 0.866) = 0.25$. Ratio: $0.75/0.25 = 3$. ✓

ANSWER

$\alpha = \dfrac{\sqrt{3}}{2} \approx 0.866$. The distribution $f(x) = 2x$ is the Beta(2,1), concentrated toward 1. The point 0.866 splits it so that 75% lies below and 25% above.

QUESTION 2

Find t such that $P(X < t) = 0.5$.

$f(x) = e^{-x}$ for $0 \leq x < \infty$ and $f(x) = 0$ otherwise.

1. **Identify the distribution.** This is the Exponential distribution with rate $\lambda = 1$. Its mean is $E[X] = 1/\lambda = 1$. We are asked for the *median*, which differs from the mean when the distribution is skewed.

2. **Derive the CDF.** Using the antiderivative $\int e^{-x}\,dx = -e^{-x} + C$:

$$F(t) = \int_0^t e^{-x}\,dx = \left[-e^{-x}\right]_0^t = -e^{-t} - (-e^0) = 1 - e^{-t}$$

3. **Set $F(t) = 0.5$ and solve.**

$$1 - e^{-t} = 0.5 \implies e^{-t} = 0.5 \implies -t = \ln(0.5) = -\ln 2 \implies t = \ln 2$$

4. **Numerical value.** $t = \ln 2 \approx 0.6931$. The median is approximately 0.693, while the mean is 1. The distribution is right-skewed: a long tail to the right pulls the mean above the median.

ANSWER

$t = \ln 2 \approx 0.693$. General result: the median of $\text{Exp}(\lambda)$ is always $\ln 2/\lambda$.

LOG ALGEBRA TO MEMORISE

$\ln(1/2) = \ln(2^{-1}) = -\ln 2$. Therefore $-\ln(0.5) = -\ln(1/2) = \ln 2$. When you take logs of an equation involving 0.5 or 1/2, expect $\ln 2$ to appear.

QUESTION 3

Find $E(X)$ and $\text{Var}(X)$ for the softball player's hit distribution.

$P(X=1)=0.3,\ P(X=2)=0.2,\ P(X=0)=3P(X=3).$

1 **Find the missing probabilities.** Let $P(X=3)=p$; then $P(X=0)=3p$. Normalisation:

$$3p+0.3+0.2+p=1 \implies 4p=0.5 \implies p=0.125$$

So $P(X=3)=0.125$ and $P(X=0)=0.375$.

2 **Build the full computational table.**

X	$P(X)$	$XP(X)$	$X^2P(X)$
0	0.375	0.000	0.000
1	0.300	0.300	0.300
2	0.200	0.400	0.800
3	0.125	0.375	1.125
Σ	1.000	**1.075**	**2.225**

3 **Compute the expected value:**

$$E[X]=1.075 \text{ hits per game}$$

4 **Compute the variance via the shortcut:**

$$\mathrm{Var}(X)=E[X^2]-(E[X])^2=2.225-(1.075)^2=2.225-1.155625=1.069375$$

ANSWERS

$E[X]=1.075$ | $\mathrm{Var}(X)\approx 1.069$ | $\mathrm{SD}(X)\approx 1.034$

QUESTION 4 *Find $E[X]$ and $\mathrm{Var}(X)$ for contracts carried forward.*

2 old contracts. New contracts J: $P(1) = 0.1, P(2) = 0.2, P(3) = 0.5, P(4) = 0.2$. Completions C: $P(C = 2) = P(C = 3) = 0.5$. $J \perp C$. $X = 2 + J - C$.

1 **Compute $E[J]$ and $\mathrm{Var}(J)$.**

J	$P(J)$	$JP(J)$	$J^2P(J)$
1	0.1	0.1	0.1
2	0.2	0.4	0.8
3	0.5	1.5	4.5
4	0.2	0.8	3.2
Σ	1.0	**2.8**	**8.6**

$$\mathrm{Var}(J) = 8.6 - (2.8)^2 = 8.6 - 7.84 = 0.76$$

2 **Compute $E[C]$ and $\mathrm{Var}(C)$.**

$$E[C] = 0.5(2) + 0.5(3) = 2.5$$

$$E[C^2] = 0.5(4) + 0.5(9) = 6.5 \implies \mathrm{Var}(C) = 6.5 - 6.25 = 0.25$$

3 **Apply linearity of expectation:**

$$E[X] = E[2 + J - C] = 2 + E[J] - E[C] = 2 + 2.8 - 2.5 = 2.3$$

4 **Apply the variance addition theorem (requires independence):**

$$\mathrm{Var}(X) = \mathrm{Var}(2 + J - C) = 0 + \mathrm{Var}(J) + (-1)^2\mathrm{Var}(C) = 0.76 + 0.25 = 1.01$$

The constant 2 contributes zero variance. The $(-1)^2 = 1$ confirms that subtracting C adds, not subtracts, its variance.

ANSWERS

$E[X] = 2.3$ contracts | $\mathrm{Var}(X) = 1.01$ | $\mathrm{SD}(X) \approx 1.005$

QUESTION 5

Determine k; find E[X] and Var(X).

$f(x) = \frac{3}{4}(kx - x^2)$ for $0 \leq x \leq 2$ and $f(x) = 0$ otherwise.

1 **Find k via normalisation. Set $\int_0^2 f(x)\,dx = 1$:**

$$\frac{3}{4}\int_0^2 (kx - x^2)\,dx = \frac{3}{4}\left[\frac{kx^2}{2} - \frac{x^3}{3}\right]_0^2 = \frac{3}{4}\left[2k - \frac{8}{3}\right] = 1$$

$$2k - \frac{8}{3} = \frac{4}{3} \implies 2k = \frac{12}{3} = 4 \implies k = 2$$

2 **Write the complete PDF with $k = 2$ and verify non-negativity.**

$$f(x) = \frac{3}{4}(2x - x^2) = \frac{3}{4}x(2 - x)$$

For $x \in [0, 2]$: $x \geq 0$ and $2 - x \geq 0$, so $f(x) \geq 0$. ✓ The product $x(2 - x)$ equals 0 at both endpoints and peaks at $x = 1$ — a parabolic arch.

3 **Compute $E[X]$.**

$$E[X] = \frac{3}{4}\int_0^2 x(2x - x^2)\,dx = \frac{3}{4}\int_0^2 (2x^2 - x^3)\,dx$$

$$= \frac{3}{4}\left[\frac{2x^3}{3} - \frac{x^4}{4}\right]_0^2 = \frac{3}{4}\left[\frac{16}{3} - 4\right] = \frac{3}{4}\cdot\frac{4}{3} = 1$$

The distribution is symmetric about $x = 1$ on $[0, 2]$, consistent with $E[X] = 1$.

4 **Compute $E[X^2]$.**

$$E[X^2] = \frac{3}{4}\int_0^2 x^2(2x - x^2)\,dx = \frac{3}{4}\int_0^2 (2x^3 - x^4)\,dx$$

$$= \frac{3}{4}\left[\frac{x^4}{2} - \frac{x^5}{5}\right]_0^2 = \frac{3}{4}\left[8 - \frac{32}{5}\right] = \frac{3}{4}\cdot\frac{8}{5} = \frac{6}{5}$$

5 **Compute $\mathrm{Var}(X)$.**

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2 = \frac{6}{5} - 1 = \frac{1}{5} = 0.2$$

ANSWERS

$k = 2$ | $E[X] = 1$ | $\mathrm{Var}(X) = 0.2$ | $\mathrm{SD}(X) = 1/\sqrt{5} \approx 0.447$

SYMMETRY CHECK

The PDF $f(x) = \frac{3}{4}x(2 - x)$ is symmetric about $x = 1$ on $[0, 2]$: substituting $u = 2 - x$ maps $f(x)$ to $f(2 - u)$, which has the same form. By symmetry, $E[X]$ must equal the midpoint of $[0, 2]$, which is 1 — confirming our integral result without calculation.

CHAPTER VI

Pattern Recognition

Identify the question type in the first ten seconds. Match the pattern; apply the corresponding strategy.

INFINITE SUPPORT, POWER FORM

- Geometric PMF
- Apply series theorem
- Identify a and r
- Require $|r| < 1$

PRODUCT OF BINOMIALS IN NUMERATOR

- Check negative arguments → 0
- Compute each term in table
- Sum and compare to 1
- Hypergeometric pattern

UNKNOWN CONSTANT K PRESENT

- Normalise first — always
- Solve for k algebraically
- Then compute $E[X]$, Var
- Never skip this step

PIECEWISE STEP FUNCTION GIVEN

- This is a CDF
- Find all breakpoints
- Jump size = PMF value
- Verify recovered sum = 1

"FIND T SUCH THAT $P = C$"

- Set up integral equation
- Evaluate CDF at t
- Solve algebraically
- Check domain bounds

SOME PROBABILITIES GIVEN, SOME AS RELATIONS

- Set unknown = variable
- Express relations
- Use $\sum p = 1$
- Solve before E[X]

X IS A FUNCTION OF INDEPENDENT RVS

- Decompose $X = f(J, C)$
- Apply linearity for $E[X]$
- Add variances (independence)
- Constants add nothing to Var

POLYNOMIAL PDF, FIND E AND VAR

- Find k first
- Build table: $xf(x)$, $x^2f(x)$
- Integrate term by term
- Use shortcut for Var

CHAPTER VII

Quick Reference

All formulae on one page. Verify you can state and apply each without consulting notes.

CONCEPT	DISCRETE	CONTINUOUS
PMF / PDF validity	$p(x) \geq 0; \sum p(x) = 1$	$f(x) \geq 0; \int f(x)\,dx = 1$

CONCEPT	DISCRETE	CONTINUOUS
CDF definition	$F(x) = \sum_{t \le x} p(t)$	$F(x) = \int_{-\infty}^{x} f(t)\,dt$
PMF/PDF from CDF	$p(x_0) = F(x_0) - F(x_0^-)$	$f(x) = F'(x)$
Expected value	$\sum x\,p(x)$	$\int x\,f(x)\,dx$
$E[X^2]$	$\sum x^2\,p(x)$	$\int x^2\,f(x)\,dx$
Variance (shortcut)	$E[X^2] - (E[X])^2$	
Standard deviation	$\sigma = \sqrt{\text{Var}(X)}$	
Linearity of E	$E[aX + b] = aE[X] + b$	
Var under scaling	$\text{Var}(aX + b) = a^2\text{Var}(X)$	
Var of indep. sum	$\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y)$	
Geometric series	$\sum_{i=0}^{\infty} ar^i = \frac{a}{1-r}, \; \lvert r \rvert < 1$	
Exponential CDF	$F(x) = 1 - e^{-\lambda x}$; median $= \frac{\ln 2}{\lambda}$	
Antiderivative	$\int e^{-x}\,dx = -e^{-x} + C$	
Key log identity	$-\ln(1/2) = \ln 2 \approx 0.693$	

Pre-Examination Checklist

- ✓ State both conditions for a valid PMF from memory
- ✓ State both conditions for a valid PDF from memory
- ✓ Apply the geometric series formula to a sum of the form $\sum ar^x$
- ✓ Evaluate $\binom{n}{k} = 0$ correctly when $k < 0$
- ✓ Recover a PMF from a step CDF using the jump-size method
- ✓ Find an unknown constant k from the normalisation condition (discrete and continuous)
- ✓ Distinguish \(P(X
- ✓ Evaluate $\int_a^b 2x\,dx$, $\int_a^b e^{-x}\,dx$, $\int_a^b (kx - x^2)\,dx$
- ✓ Set up and solve a threshold problem of the form $P(X < \alpha) = c$

- ✓ Solve $e^{-t} = 0.5$ using natural logarithms
- ✓ Find missing probabilities from a partial PMF specification
- ✓ Construct the $x,\, p(x),\, x \cdot p(x),\, x^2 \cdot p(x)$ table correctly
- ✓ Compute $E[X]$ and $E[X^2]$ from the table and apply the shortcut variance formula
- ✓ Apply linearity of expectation to decompose a compound RV
- ✓ Add variances of independent RVs, including for the case $X - Y$